

Planisuss Project Report

Introduction



The Planisuss project is designed to simulate a simplified and interactive ecosystem (with three species, each has their own behavior). The aim is to accurately model the interactions and behaviors of these three species within a grid-based environment, thereby providing insights into ecosystem dynamics.

1. Goals of the Project

a. Problems to be Solved

i. Species Interaction

Modeling realistic interactions between different species, such as predation, grazing, and social behavior.

ii. Resource Management

Simulating the availability and consumption of resources within the ecosystem.

iii. Movement and Decision-Making

Implementing decision-making processes for individual animals and groups based on their environment and internal states.

iv. Simulation Performance

Efficiently simulating and visualizing a large number of entities within a reasonable time frame.

b. Constraints

i. Computational Resources

The simulation must run efficiently on standard computational hardware, ensuring accessibility and usability for a wide range of users.

ii. Realistic Behavior

The behavior of species should be modeled realistically, to accurately reflect natural behaviors and interactions. The simulation must provide clear and informative visual feedback to the user.

2. Assumptions, Methods, Procedures

a. Motivation of Design Choices

i. Grid-Based World

Using 2 dimensional grids simplifies spatial. It provides a clear and structured framework for modeling the ecosystem.

ii. Discrete Time Steps

The simulation progresses in discrete time increments, with each step representing one day. This allows for clear and manageable tracking of changes over time.

iii. Object-Oriented Programming

The choice of object-oriented programming makes the entire structure clearer and allows for easy addition of new species and behaviors without altering the overall structure of the simulation.

iv. Randomization

Incorporating randomization adds variability and realism to the simulation. It ensures that each run is unique, reflecting the unpredictable nature of real-world ecosystems.

v. Memory and Social Behavior

Integrating memory and social behavior into decision-making processes enhances realism.

Animals make decisions based on past experiences and social interactions, mimicking real-life behaviors.

b. Logical Structure of the Solution

i. Initialization

The world grid is initialized by setting up boundaries, water cells, and the initial populations of various species.

ii. Daily Actions

Each day in the simulation, species perform actions such as growing, moving, grazing, and predating based on their predefined behaviors.

iii. Logging and Visualization

The simulation logs population data and visualizes the state of the world at each time step, to track changes, analyze patterns, and gain insights into the ecosystem's dynamics.

iv. User Interaction

Users can interact with the simulation through key presses to pause, continue, reset, or close the simulation.

c. Implementation

i. Classes

1. Main Simulation Class (Planisuss)

Manages the world grid, species populations, and daily actions. Handles initialization, daily actions, logging, and user interactions.

2. Species Classes (Vegetob, Erbast, Carviz)

Represent different individual species with specific behaviors and attributes. Define the growth, movement, grazing, and predation behaviors of each species.

3. Group Classes (Herd, Pride)

Represent groups of animals with collective behaviors. Model the formation, movement, and interactions of groups.

4. Visualization and Logging

Handle the visualization of the grid and logging of population data. Provide real-time visual feedback and record data for analysis and understanding of the simulation's dynamics.

ii. Decision-Making and Behavior of the Three Species

1. Vegetob (Plant)

Vegetob is stationary and focuses on growth. Each day, it increases its density up to a maximum of 100. It serves as a resource for Erbast and Herds. If its density reaches zero due to grazing, it is removed from the grid.

2. Erbast (Herbivore)

Erbast graze on Vegetob to increase their energy. They can form Herds if there are multiple Erbast in the same cell. Erbast and Herds move based on their social attitude and environmental conditions. Each day, Erbast decide whether to move based on the desirability of surrounding positions, considering factors like predator presence, availability of Vegetob, and previously visited positions. Erbast graze on Vegetob, reducing its density. They avoid predators like Carviz. When grouped into Herds, they move and graze collectively, increasing efficiency and survival chances.

3. Carviz (Carnivore)

Carviz hunt Erbast and Herds to increase their energy. They can form Prides if there are multiple Carviz in the same cell. Carviz and Prides move based on their social attitude and environmental conditions. Each day, Carviz decide whether to move based on the desirability of surrounding positions, considering factors like prey availability, predator presence, and previously visited positions. Carviz hunt Erbast and Herds, increasing their energy and reducing the prey population. When grouped into Prides, they hunt collectively, increasing their hunting success rate. When multiple Prides meet, they engage in fight.

iii. Daily Workflow in Planisuss

1. Growth of Plants

Vegetob entities grow by increasing their density up to a maximum value. If surrounded by maximum density Vegetob, Erbastas at that position are overwhelmed and removed.

2. Grouping of Animals

- Erbastas into Herds: Multiple Erbastas in the same position group into Herds, enhancing their grazing efficiency and survival.
- Carviz into Prides: Multiple Carviz in the same position group into Prides, improving their hunting efficiency and social interactions.

3. Movement

- Herds: Herds decide whether to move based on the desirability of surrounding positions, considering factors of past memories, availability of food and presence of predators. If a new position is chosen, the Herd moves to that position, consuming a small amount of energy for each member. If the group doesn't have enough collective energy to move, it stays in its current position.
 - Memory: Positions in the group's memory receive a negative score to encourage exploration and avoid overcrowding.
 - Food Sources: Presence of Vegetob increases a position's score.
 - Predators: Presence of Carviz and Prides significantly reduces a position's score.
- Prides: Prides decide whether to move based on similar factors as Herds, but with a focus on prey availability.
- Erbastas: Erbastas not in Herds decide whether to move based on their social attitude and environmental factors, similar to Herds.
- Carviz: Carviz not in Prides decide whether to move based on their social attitude and environmental factors, similar to Prides.

4. Grazing and Predation

- Grazing: Erbastas and Herds which did not choose to move graze on Vegetob, increasing their energy and reducing Vegetob density. Herds prioritize feeding based on the energy levels of individual members, ensuring those with the least energy feed first. If Vegetob density reaches zero, it is removed from the grid.
- Predation: Carviz and Prides hunt Erbastas and Herds, increasing their energy and reducing the prey population if the hunt is successful.
 - Prides look for prey within their current position and prioritize targets based on energy levels, typically going after the most energy-rich prey available.

- The success of a hunt is determined by the collective energy of the pride compared to the difficulty of the prey. A higher collective energy increases the pride's success probability, making successful hunts more likely.
- When a pride decides to hunt, it attempts to capture the identified target. If successful, the pride's members share the prey's energy, increasing their individual energy levels.

5. Conflict Resolution

- Herds: Multiple Herds in the same position resolve conflicts by merging into a single Herd.
- Prides: Multiple Prides in the same position resolve conflicts either by joining (based on social attitude) or fighting. Prides with a higher combined social attitude are more likely to join forces, while those with lower attitudes may fight for dominance. If the prides decide to join, they merge their members, increasing the group's size and strength. If they decide to fight, the prides engage in combat, and members with lower energy are removed until one pride emerges victorious. The winner absorbs the members of the losing Pride.

6. Spawning

The age value of the animals are increased by one day. Those that reach an age multiple of 10 (one month) have their energy decreased by aging. Animals that reach max age spawn two offsprings, with age set to 0, distributing their energy and other properties among the new entities.

7. Removal of Dead Entities

Entities with zero energy or that have reached their lifetime are removed from the simulation.

8. Logging and Visualization

The state of the simulation is logged, including species positions and counts. The current state of the grid is visualized, showing different species with specific colors and providing interactive controls for the user.

3. Tools and Resources

a. Libraries Used

- Python 3.10: The primary programming language used for the simulation.
- NumPy Version 1.21.0: Used for numerical operations and managing the grid.
- Matplotlib Version 3.4.2: Used for visualization of the simulation.
- Random (part of the Python standard library): Used for random number generation to simulate behaviors.

v. Pickle (part of the Python standard library): Used for saving and loading the simulation state.

b. References

Wa-tor

Conway's Game of Life

"Planisuss — Final exam project — A.Y. 2022/23"

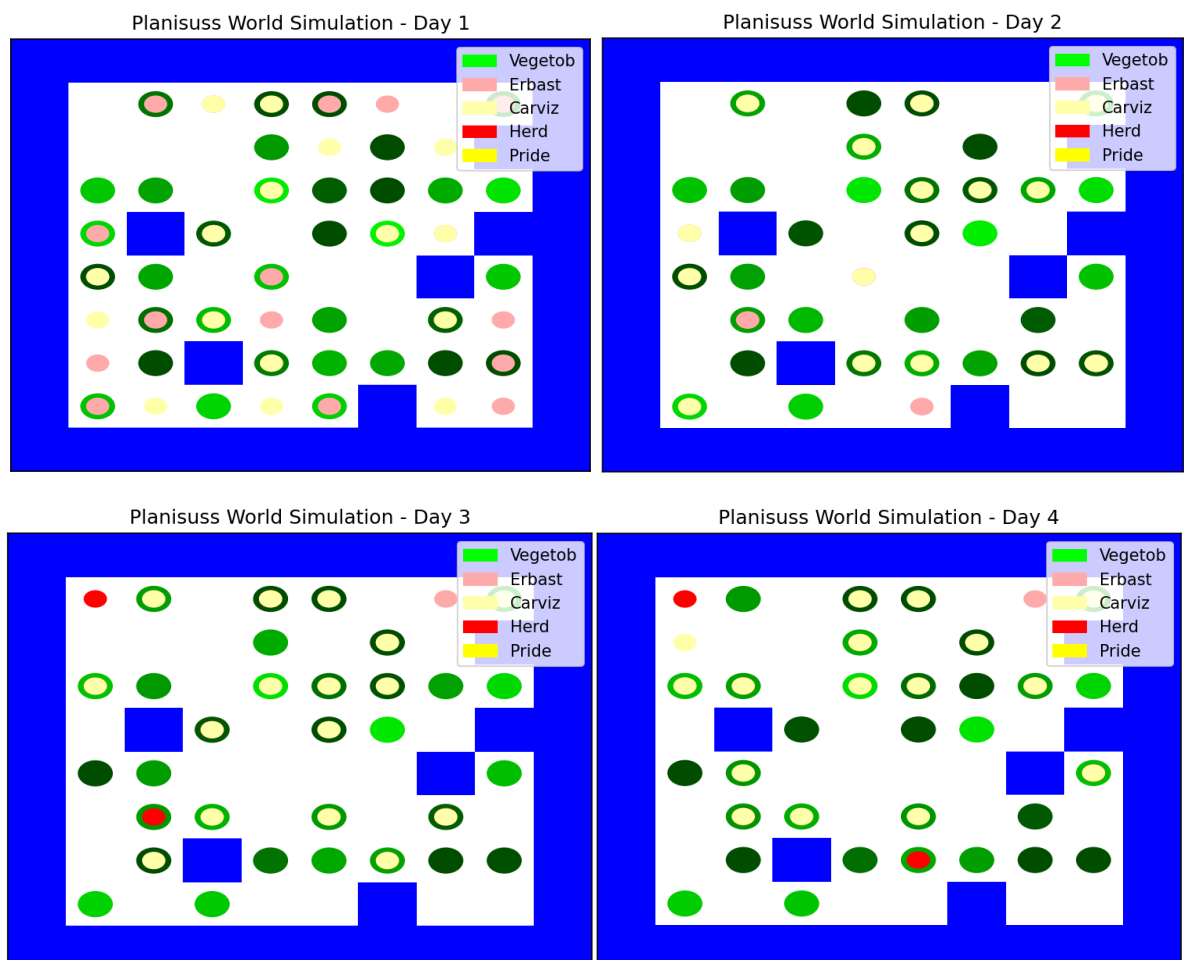
4. Testing and Results

The Planisuss simulation provides a comprehensive visualization of the grid and species interactions.

a. Grid Visualization

The grid is displayed with different colors representing various entities:

- Water: Blue cells
- Vegetob: Shades of green, varying with density
- Erbast: Light red circles
- Carviz: Light yellow circles
- Herd: Red circles
- Pride: Yellow circles
- Each species and group type is clearly distinguishable by color and shape.



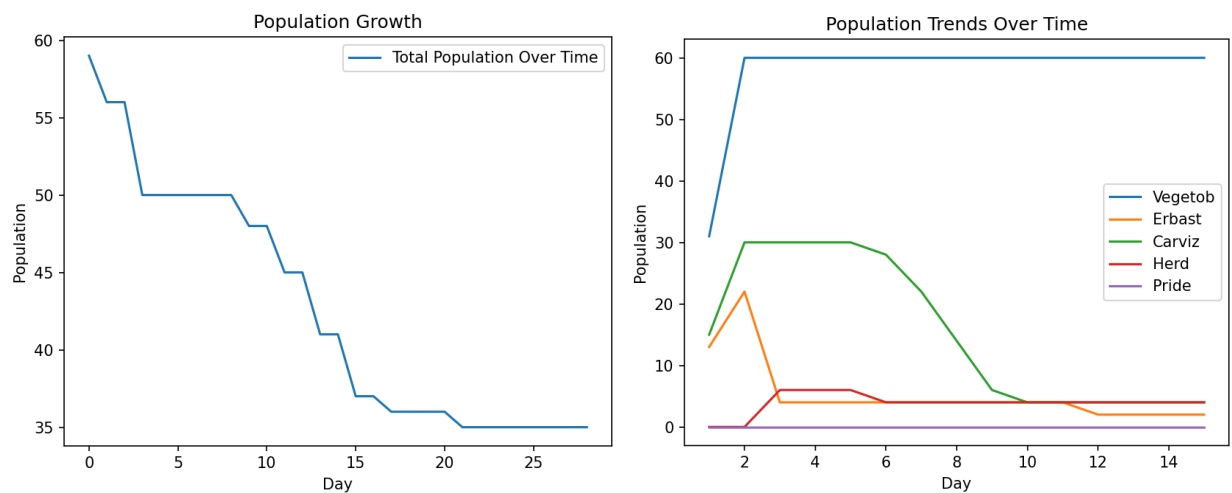
b. Interactive Controls

The visualization includes interactive controls allowing the user to:

- Pause/Continue: Pause or continue the simulation.
- Reset: Reset the simulation to the initial state.
- Close: Close the simulation visualization.

c. Population and Trajectory Plots

- Population Growth Plot: Displays the total population over time.
- Herd Trajectories Plot: Shows the movement trajectories of Herds over time.
- Cell Details Plot: Provides detailed data for a specific cell, including Vegetob density and Erbast count.
- Population Trends Plot: Displays the population trends of different species over time.



d. Logging

The simulation logs daily snapshots, capturing the state of the world, species positions, and counts. This data is used for plotting population trends and analyzing the simulation's dynamics.

5. Possible Improvements

a. Additional Species

Introduce more species with unique behaviors and interactions to enhance the complexity and realism of the simulation.

b. Resource Dynamics

Implement more complex resource dynamics, such as regeneration and depletion of vegetation over time.

c. Enhanced Decision-Making

Improve the decision-making algorithms for animals and groups to consider additional factors, such as long-term survival strategies and cooperative behaviors.

d. Parallelization

Optimize the simulation to run in parallel, distributing computational tasks across multiple processors.

e. User Interface

Develop a graphical user interface (GUI) to provide more intuitive controls and enhanced visualizations.

Conclusion

The Planisuss project successfully models the interactions between species within a grid-based world, providing a dynamic and interactive ecological simulation. The detailed implementation, including the use of object-oriented design, randomization, memory, and social behavior, ensures realistic behavior and interactions. The visualization and logging capabilities allow users to observe and analyze the evolution of the simulated ecosystem. Future improvements can enhance the complexity, realism, performance, and user experience of the simulation.